Chapter 3: Sample drawing Techniques

1. Venipuncture

Definition: Venipuncture refers to the procedure of obtaining intravenous access for intravenous therapy or blood withdrawal. Venipuncture is simply achieved with a butterfly needle, vacuum tube, or syringe. The blood consists of plasma, red blood corpuscles, white blood corpuscles, and platelets. The function of transporting oxygen, nutrients, and body wastes is performed by the blood. The blood samples are used in every clinical laboratory on a regular basis for CBC (Complete Blood Count), tests of liver and kidneys, and electrolyte analysis.

Procedure:

1.Patient Preparation: Verify patient identification, educate the patient, and get consent.

2.Site Selection: The sites are usually chosen in the median cubital vein, cephalic vein, or basilic vein.

3.Aseptic Technique: Clean with alcohol swabs and allow it to dry.

4. Need Insertion: Insert the needle 15-30°, aspirate for blood, and take out the needle when obtaining the sample.

5. Post-procedure Care: Squeeze gauze over the site, label the specimen, and dispose of sharps.

Complications:

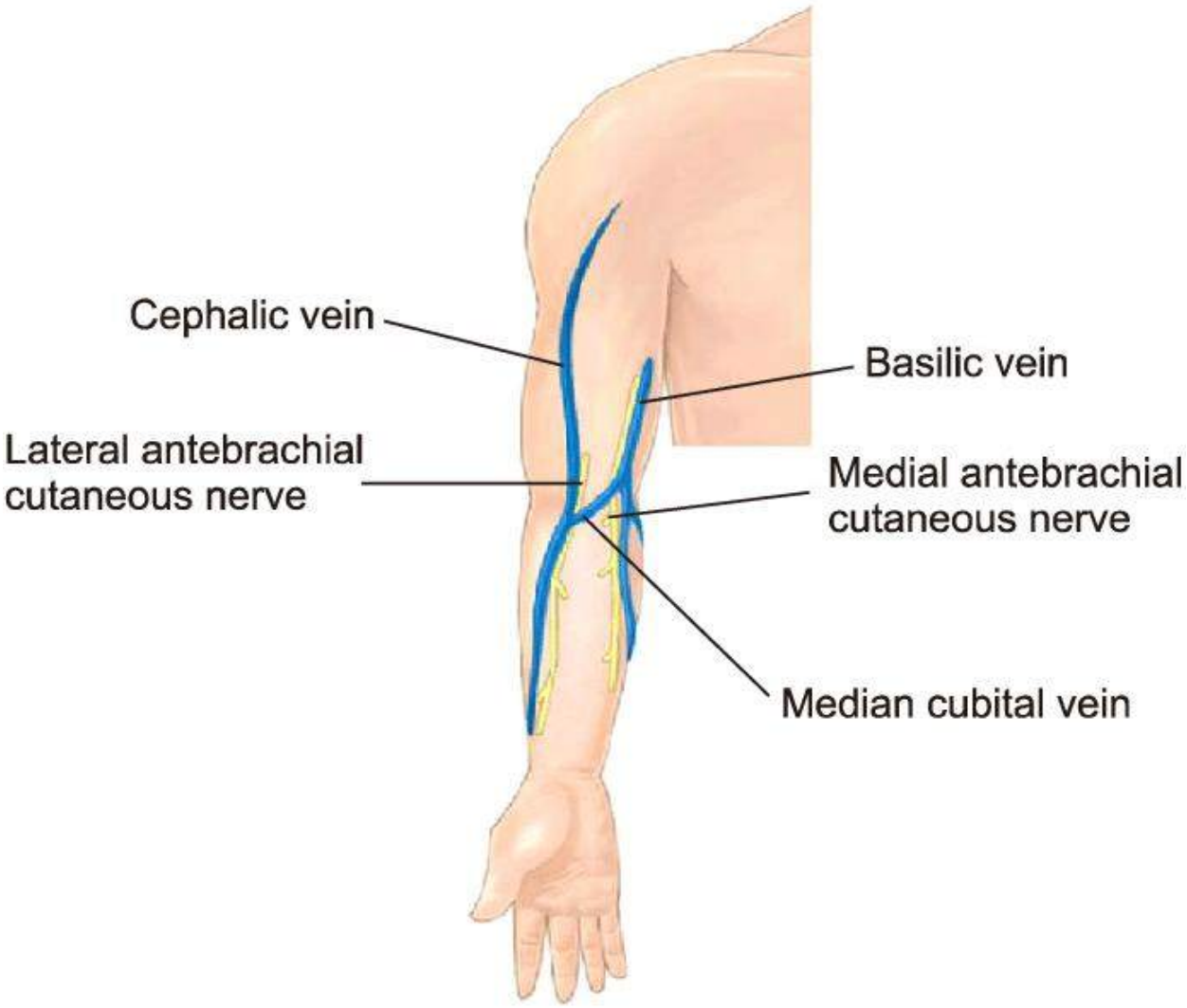
• Hematoma formation

• Phlebitis

• Nerve damage

• Infection

https://www.youtube.com/watch?v=DzTmDlcE_3U



Arterial Blood Gas:

Arterial Blood Gas (ABG) analysis is an important test for the evaluation of a patient's oxygenation, ventilation, and acid-base balance. The radial artery is frequently selected for sampling because of its accessibility and collateral circulation.

Procedure for Radial Artery Blood Sampling:

1.Patient Preparation:

- **Identification:** Verify the patient's identity and describe the procedure to obtain informed consent.

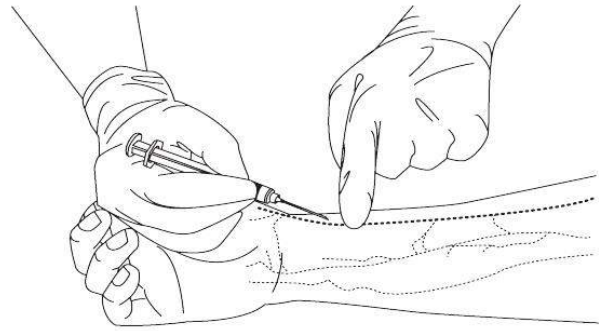
- **Positioning:** Place the patient in a supine position with the wrist extended and supported by a rolled towel.

2.Site Preparation:

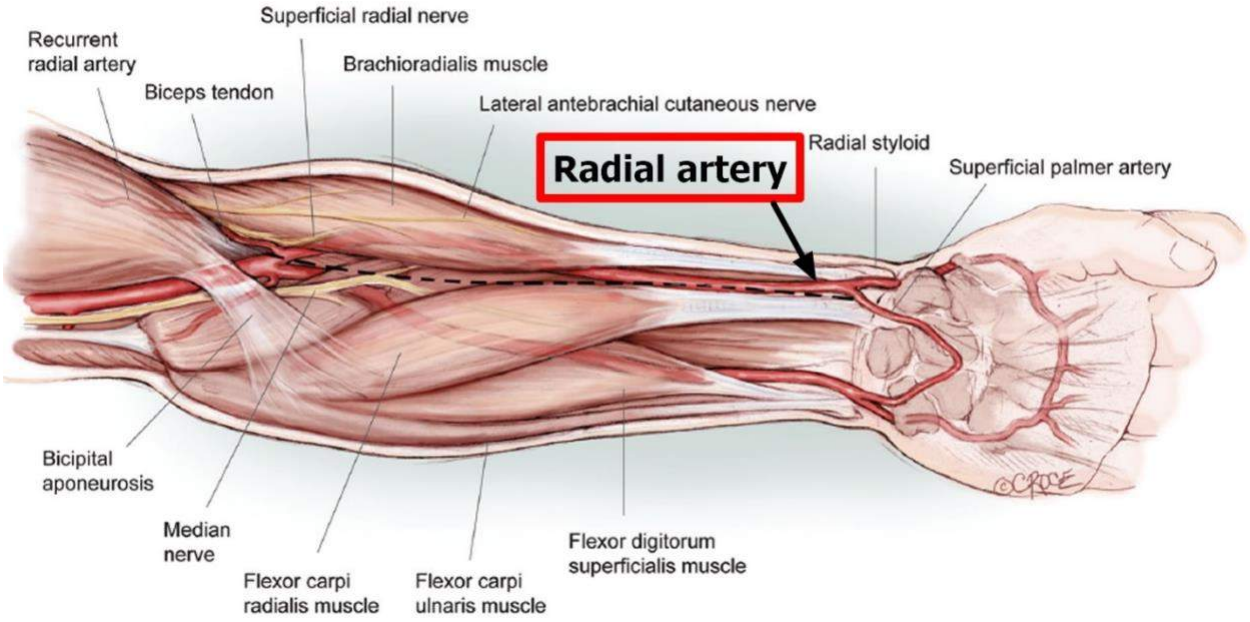
- Perform hand hygiene and don non-sterile gloves.
- Clean the puncture site with 70% isopropyl alcohol and let air dry.

3.Sample Collection:

- **Equipment:** Use a pre-heparinized syringe with 20–25 gauge needle.
- **Technique:**
    1. Locate the pulse of the radial artery by palpation
    2. Insert the needle at the 45-degree angle, bevel up, distal to the palpating fingers.
    3. Allow the syringe to fill spontaneously; do not pull back on the plunger.
    4. Once the desired volume (usually 1–2 mL) is achieved, withdraw the needle immediately



Locate artery and take a sample



4.Post-Procedure Care:

- Use firm pressure to the puncture site immediately with sterile gauze for at least 5 minutes to prevent hematoma formation.
- When bleeding is under control, bandage the site with a sterile dressing.

5.Sample Handling:

- Expel air bubbles from the syringe to prevent alteration of gas values.
- Cap the syringe and agitate the sample gently to determine that it is properly anticoagulated.
- Label the sample appropriately with the patient identification, and transport the sample to the laboratory immediately, preferably being analyzed within 10-15 minutes if stored at room temperature.

Complication

1. Haematoma or excessive bleeding
2. Nerve damage
3. Fainting or a vasovagal response can be prevented by ensuring that the patient is supine (lying down on their back) with feet elevated before beginning the blood draw
4. https://www.youtube.com/watch?v=0BSv4iN8T2E

Capillary Sampling

Introduction:

Capillary blood sampling is a minimally invasive technique that is utilized when only a small quantity of blood is required, e.g., for glucose testing, hemoglobin measurement, or newborn screening. It is extremely helpful in children and elderly patients or where venous access is difficult. Capillary sampling involves the removal of blood from small vessels close to the skin surface, most commonly from the fingertip, heel, or earlobe.

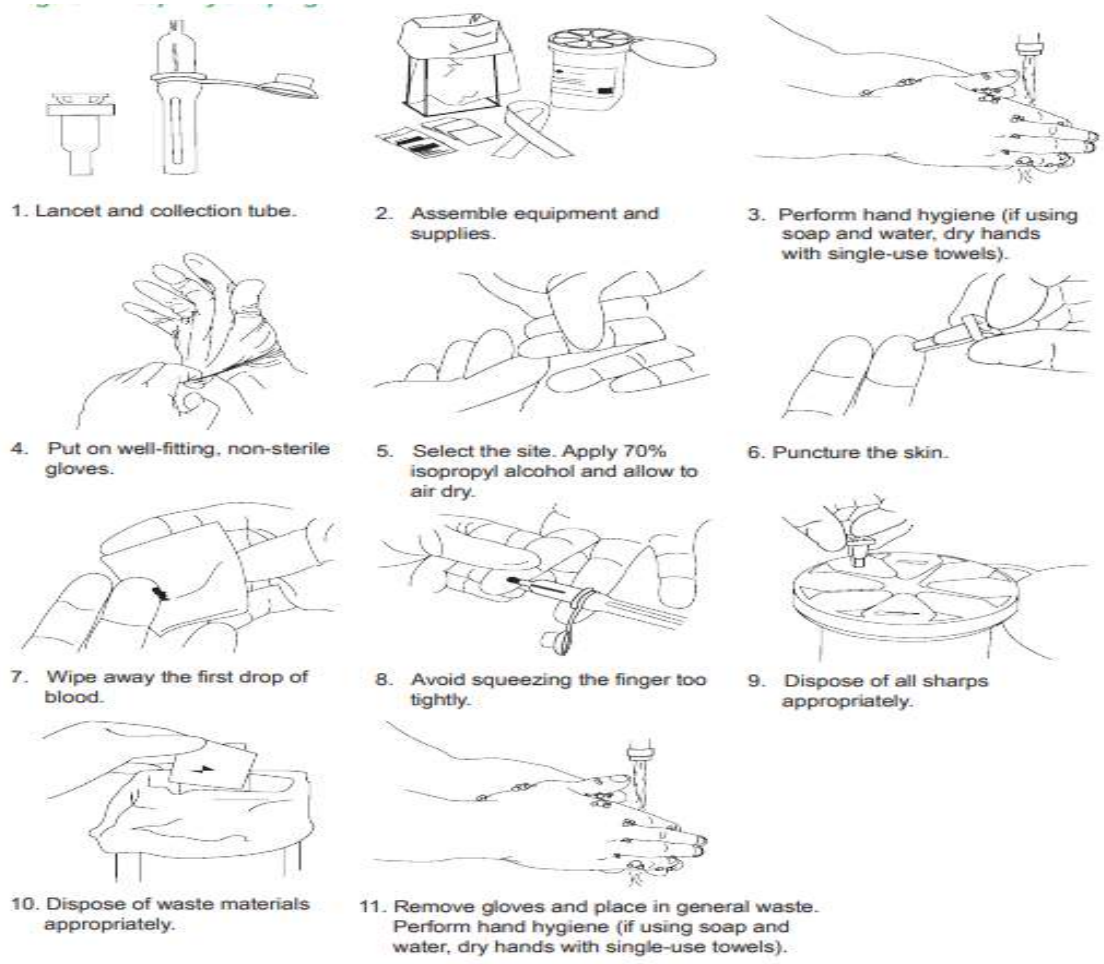
Procedure for Capillary Blood Sampling

1. Patient Preparation:

- Identify and explain the procedure to the patient to obtain their cooperation.
- Select a suitable puncture site:
    1. Adults & older children: Fingertip (middle or ring finger)
    2. Neonates & infants: Heel (medial or lateral plantar surface)
    3. Avoid sites which have been punctured before, swollen, infected, or bruised.

2. Site Preparation:

- Wear gloves and wash your hands.
- Prepare the chosen site with 70% isopropyl alcohol and let it air dry.
- Warm the site (if necessary) with a warm compress to increase blood flow.



3. Sample Collection:

- Use a sterile lancet or an automated skin puncture device.
- For fingertip sampling, puncture transverse to the fingerprint lines to allow proper flow.
- For neonatal heel sampling, puncture the appropriate heel area to avoid injury.
- Blot away the first drop of blood to remove tissue fluid contamination.
- Permit blood to enter the collection tube naturally without too much squeezing to avoid dilution with interstitial fluid.

4. Post-Procedure Care:

- Use sterile gauze to control bleeding.
- Dispose of the lancet in a sharp's container.
- Label the sample appropriately and transport it immediately.

Capillary sampling is a valuable technique for quick and easy blood sampling, minimizing patient discomfort while providing accurate biochemical analysis.

https://www.youtube.com/watch?v=M025dqNgB-0

CSF Collection

Cerebrospinal fluid (CSF) tapping, or lumbar puncture, is a medical procedure used to obtain a sample of the fluid surrounding the brain and spinal cord. The fluid serves as a cushion to protect these structures from injury, and lab testing can aid in the diagnosis of certain neurological illnesses.

Procedure for CSF Collection:

1.Preparation:

- **Patient Position:** The patient is asked to lie on one side with the knees folded into the chest and chin down toward the chest. Alternatively, the procedure can be administered in a sitting position leaning forward.
- **Site Preparation**: The region where the lower spine is situated is disinfected with antiseptic for a sterile environment.

2.Anesthesia:

- A local anesthetic medication is applied at the lower spinal area to insensitize the site, thus alleviating pain experienced during treatment.

3.Needle Insertion

- A spinal needle is inserted gently between the lumbar vertebrae into the subarachnoid space containing the CSF.
- In certain situations, imaging guidance, such as fluoroscopy, may be employed to verify correct placement of the needle.

4.CSF Collection:

- The needle is left in place and a small amount of CSF (usually 1 to 10 milliliters) is aspirated into sterile containers for examination.

5.Post-Procedure:

- The needle is withdrawn and the puncture site is dressed.

- Patients are typically instructed to lie flat for a brief time to minimize the risk of headache and are advised to take more fluids after the procedure.

Analysis of CSF can be of immense assistance in the diagnosis of such conditions as infections (e.g., meningitis), hemorrhage around the brain (subarachnoid hemorrhage), and other neurologic disorders.

https://www.youtube.com/watch?v=9kckrlBIR2E